$$-2 \begin{bmatrix} 1 & 3 & -4 \\ 0 & 5 & 2 \end{bmatrix} = \begin{bmatrix} -2 & -6 & 8 \\ 0 & -10 & -4 \end{bmatrix}$$

The scalar reaches every entry alike: each one is multiplied by −2, and the shape and the zero stay where they were.